Theory of Value

An Axiomatic Analysis of Economic Equilibrium
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Chapter 1: Mathematics

01. Introduction
02. Sets
03. Functions and Correspondences
04. Preorderings
05. Real Numbers
06. Limits in $\mathbb{R}^N$
07. Continuous Functions
08. Continuous Correspondences
09. Vectors in $\mathbb{R}^N$
10. Fixed Points

Preorderings

Let $\mathscr{R}$ be a binary relation in which any two elements x_1 and x_2 of S (the order of which is essential) stand or do not stand. If they do, one writes $x_1 \mathscr{R} x_2$.

Example: (1) Let $\mathscr{S}$ be the set of subsets of a set S. The relation $\mathscr{R}$ on $\mathscr{S}$ might be "is contained in," then $X \mathscr{R} X'$ would be equivalent to $X \subseteq X'$: (2) or the relation $\mathscr{R}$ might be "does not intersect," then $X \mathscr{R} X'$ would be equivalent to $X \cap X' = \varnothing$. (3) Let A be the set of actions available to an agent; the relation $\mathscr{R}$ might be "is not preferred to." (4) Consider the relation "is not a successor of" on $\mathbb{N}$ and denote it by $\leqq$.

Preorderings

The last relation corresponds to the natural ordering of the elements of $\mathbb{N}$. To define with full generality an ordering relation on a set, one preserves certain properties of $\leq$ on $\mathbb{N}$. In a precise manner, a binary relation $\mathscr{R}$ on S which satisfies

(1) $x\mathscr{R}x$ for every $x \in S$ (reflexivity),

(2) "$x_1\mathscr{R}x_2$ and $x_2\mathscr{R}x_3$" implies "$x_1\mathscr{R}x_3$" (transitivity)

is called a preordering (often also a quasi-ordering). When, in addition, "$x_1\mathscr{R}x_2$ and $x_2\mathscr{R}x_1$" implies "$x_1 = x_2$", the relation is called an ordering. Often the symbols $\precsim$ will be used (in place of $\mathscr{R}$) to denote a preordering. By definition, $x_2 \succsim x_1$ means $x_1 \precsim x_2$.

Example: In the examples of 1.4.a, the first and the fourth relation are preorderings and (in fact, they are orderings), the second is not. The third will always be assumed below to be a preordering: "a^1 is not preferred to a^2" will be denoted $a^1 \precsim a^2$, this relationship is not necessarily an ordering.

Preorderings

"$x_1 \precsim x_2$ and $x_2 \precsim x_1$" is written "$x_1 \sim x_2$". "$x_1 \precsim x_2$ and not $x_2 \precsim x_1$" is written "$x_1 \prec x_2$".

Example: In the last example, $a^1 \sim a^2$ is read "a^1 is indifferent to a^2" and $a^1 \prec a^2$ is read "a^1 is less desired than a^2" or "a^2 is preferred to a^1"

Preorderings

Two elements x_1 and x_2 of a preordered set may not be comparable (as the example of the preordering $\subseteq$ emphasizes). When one necessarily has $x_1 \precsim x_2$ or $x_2 \precsim x_1$ (or both), the preordering is called complete (to emphasize that a certain is not necessarily complete, one calls it partial). Then $x_1 \precsim x_2$ is the negation of $x_1 \succ x_2$.

Example: The preordering "is not preferred to" on the set of actions available to an agent will always be assumed below to be complete.

Preorderings

Figure 1 summarizes the different types of preorderings. An arrow indicates decreasing generality.

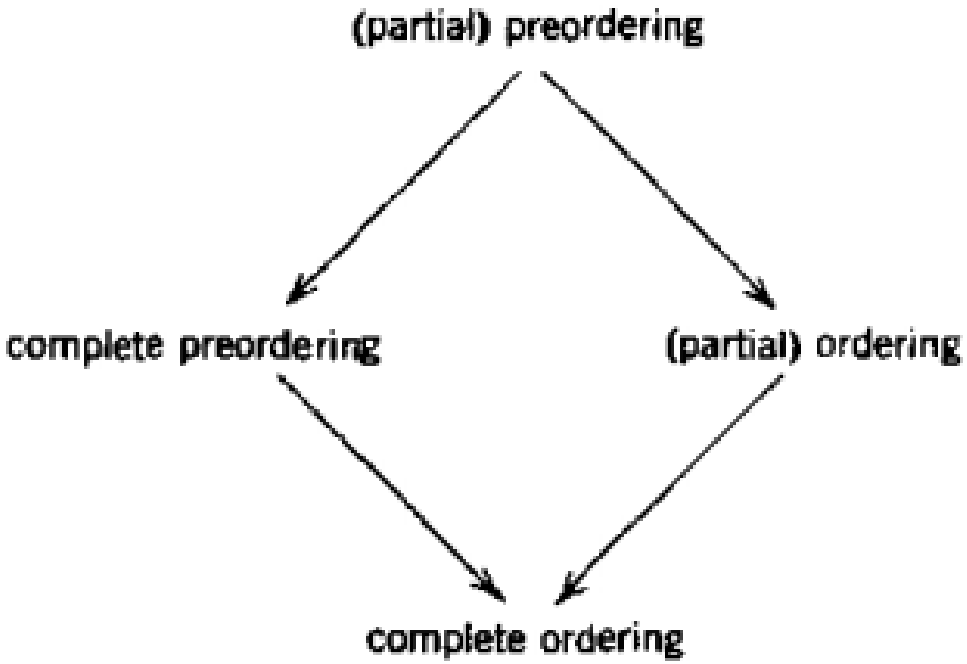


Fig. 1

Preorderings

Let S be a partially preordered set. When $y \in S$ and there is no $x \in S$ such that $x \succ y$ (resp. $x \prec y$), y is called a maximal (resp. minimal) element of S. When $y \in S$ and for all $x \in S$ one has $x \precsim y$ (resp. $x \succsim y$), y is called a greatest (resp. least) element of S. When the preordering is complete the converse is also true and the distinction disappears. When the preordering is an ordering there is at most one greateat element and at most one least element.

Example: (1) Consider the set $\mathbb{N}$ partially ordered by the relation "is a divisor of," and let S be $\{1, 2, 3, 4, 5\}$. S has three maximal elements, 3, 4, 5, no greatest element, one minimal element, 1, and one least element, 1. (2) Consider a set A of actions completely preordered by the relation "is not preferred to." If a is an action to which none is preferred, a is a greatest element of A (which need not be unique: there may be other greatest elements, all indifferent to a).

Preorderings

Let S be a preordered set, and consider a subset X of S. An element $y \in S$ such that for all $x \in X$ one has $x \precsim y$ (resp. $x \succsim y$) is called an upper bound of X (resp. a lower bound of X). Consider the set Y of the upper bounds of X; the set Y has the property: "$y \in Y$ and $y' \succsim y$" implies "$y' \in Y$." A least element of Y is a least upper bound of X. If X has a greatest element y, y is clearly a least upper bound. Consider the set Y' of the lower bounds of X; the set Y' has the property: "$y \in Y'$ and $y \succsim y'$" implies "$y' \in Y'$." A greatest element of Y' is a greatest lower bound of X. If X has a least element y, y is clearly a greatest lower bound.

Preorderings

In the next section, the requirement will be put on a certain ordered set that every non-empty subset which has an upper bound has a least upper bound.

Preorderings

Let S be a preordered set. A subset I of S is an interval if "$x_1 \in I$, $x_2 \in I$ and $x_1 \precsim x \precsim x_2$" implies "$x \in I$." Let a, b be two elements of S such that $a \precsim b$; particular cases of intervals are:

$$[a,b] = \{x \in S \mid a \precsim x \precsim b\}, \qquad (a,b) = \{x \in S \mid a \prec x \prec b\},$$
$$[a,b) = \{x \in S \mid a \precsim x \prec b\}, \qquad (a,b] = \{x \in S \mid a \prec x \precsim b\},$$
$$[a,\rightarrow) = \{x \in S \mid a \precsim x\}, \qquad (\leftarrow,b] = \{x \in S \mid x \precsim b\}.$$

Preorderings

Denote by $S_1, \cdots, S_n, \cdots, S_N$ N preordered sets, by $\precsim_n$ the preordering on S_n, by x_n a generic element of S_n. A preordering $\precsim$ is defined on the product $S = \prod_{n=1}^{N} S_n$ by $(x_n) \precsim (x'_n)$ if $x_n \precsim_n x'_n$ for every $n\,(= 1, 2, \cdots, N)$. According to the general notation of 1.4.c, $(x_n) \prec (x'_n)$ means that (α) for all n, $x_n \precsim_n x'_n$ and (β) not for all n, $x'_n \precsim_n x_n$, i.e., (β) for at least one n, $x_n \prec_n x'_n$. The notation $(x_n) \precnsim (x'_n)$ will express that, for all n, $x_n \prec_n x'_n$. With n the exception of trivial cases, the preordering $\precsim$ on S cannot be complete.

Preorderings

Let S and T he two preordered sets, and denote by $\precsim_S$ (resp. $\precsim_T$) the preordering on S (resp. on T). A function f from S to T is said to be increasing (or to be a representation of S in T) if "$x \sim_S x'$" implies "$f(x) \sim_T f(x')$" and "$x \prec_S x'$" implies "$f(x) \prec_T f(x')$".

Binary Relations

For a given set S, let $S \times S$ denote the usual Cartesian product of all ordered pairs (x_1, x_2), where both x_1 and x_2 are from S. A *binary relation* $\mathscr{R}$ on the set S is formally defined as a subset of $S \times S$ — Write $\mathscr{R} \subseteq S \times S$, and $(x_1, x_2) \in \mathscr{R}$ if the ordered pair (x_1, x_2) is in the relation $\mathscr{R}$. Another, quickly way to write $(x_1, x_2) \in \mathscr{R}$ is $x_1 \mathscr{R} x_2$, which can be read as "x_1 stands in the relation $\mathscr{R}$ to x_2." If $(x_1, x_2) \notin \mathscr{R}$, We'll write "$x_1 \not\mathscr{R} x_2$".

Binary Relations

A binary relation $\mathscr{R}$ on a set S is:

- *reflexive* if $x\mathscr{R}x$ for all $x \in S$;
- *irreflexive* if $x\not\mathscr{R}x$ for all $x \in S$;
- *symmetric* if $x_1\mathscr{R}x_2$ implies $x_2\mathscr{R}x_1$;
- *asymmetric* if $x_1\mathscr{R}x_2$ implies $x_2\not\mathscr{R}x_1$;
- *antisymmetric* if $x_1\mathscr{R}x_2$ and $x_2\mathscr{R}x_1$ implies $x_1 = x_2$;
- *transitive* if $x_1\mathscr{R}x_2$ and $x_2\mathscr{R}x_3$ implies $x_1\mathscr{R}x_3$;
- *negatively transitive* if $x_1\not\mathscr{R}x_2$ and $x_2\not\mathscr{R}x_3$ implies $x_1\not\mathscr{R}x_3$;
- *complete* or *connected* if for all $x_1 \in S$ and $x_2 \in S$, $x_1\mathscr{R}x_2$ or $x_2\mathscr{R}x_1$ (or both);
- *weakly connected* if for all $x_1 \in$ and $x_2 \in S$, $x_1 = x_2$ or $x_1\mathscr{R}x_2$ or $x_2\mathscr{R}x_1$;
- *acyclic* if $x_1\mathscr{R}x_2, \cdots, x_{N-1}\mathscr{R}x_N$ imply $x_1 \neq x_N$.

Binary Relations

An agent is willing to express his preferences among these items by making paired comparisons of the form: "I strictly prefer x_1 to x_2" which is written $x_1 \succ x_2$. "Strict preference" is a binary relation on S. Consider the following properties that this binary relation might possess:

(1) *Asymmetry* - if $x_1 \mathscr{R} x_2$, then $x_2 \mathscr{P} x_1$.

(2) *Transitivity* - if $x_1 \mathscr{R} x_2$ and $x_2 \mathscr{R} x_3$, then $x_1 \mathscr{R} x_3$.

(3) *Irreflexivity* - $x \mathscr{P} x$ for all $x \in S$.

(4) *Negative transitivity* - if $x_1 \mathscr{P} x_2$ and $x_2 \mathscr{P} x_3$, then $x_1 \mathscr{P} x_3$.

Binary Relations

[Lemma 1.] A binary relation $\mathscr{R}$ is negatively transitive if and only if $x_1 \mathscr{R} x_2$ implies that, for all $x \in S$, $x_1 \mathscr{R} x$ or $x \mathscr{R} x_2$.

Proof.

$$
\begin{aligned}
& \text{Binary relation } \mathscr{R} \text{ is negatively transitive} \\
\iff\ & \text{If } x_1 \mathscr{R} x \text{ and } x \mathscr{R} x_2 \text{ for some } x \in S, \text{ then } x_1 \mathscr{R} x_2 \\
\iff\ & \text{If not } x_1 \mathscr{R} x_2, \text{ then not } [x_1 \mathscr{R} x \text{ and } x \mathscr{R} x_2] \text{ for all } x \in S \\
\iff\ & \text{If not } x_1 \mathscr{R} x_2, \text{ then not } x_1 \mathscr{R} x \text{ or not } x \mathscr{R} x_2 \text{ for all } x \in S \\
\iff\ & \text{If } x_1 \mathscr{R} x_2, \text{ then } x_1 \mathscr{R} x \text{ or } x \mathscr{R} x_2 \text{ for all } x \in S.
\end{aligned}
$$

Preference Relations

A binary relation $\succ$ on a set S is called a strict preference relation if it is asymmetric and negatively transitive.

(1) $\succ$ *is asymmetric:* if $x_1 \succ x_2$, then $x_2 \not\succ x_1$.

(2) $\succ$ *is negative transitive:* if $x_1 \not\succ x_2$ and $x_2 \not\succ x_3$, then $x_1 \not\succ x_3$.

Preference Relations

[Proposition 1.] If $\succ$ is a strict preference relation, then it is irreflexive, transitive, and acyclic.

Proof. [1] Asymmetry directly implies irreflexivity.

[2] Suppose $x_1 \succ x_2$ and $x_2 \succ x_3$. By negative transitivity and Lemma 1, $x_1 \succ x_2$ implies either $x_1 \succ x_3$ or $x_3 \succ x_2$. But $x_3 \succ x_2$ is impossible because $x_2 \succ x_3$ is assumed and $\succ$ is asymmetric. Thus $x_1 \succ x_3$, which is transitivity.

Suppose $x_1 \succ x_2$ and $x_2 \succ x_3$. By negative transitivity and Lemma 1, $x_2 \succ x_3$ implies either $x_2 \succ x_1$ or $x_1 \succ x_3$. But $x_2 \succ x_1$ is impossible because $x_1 \succ x_2$ is assumed and $\succ$ is asymmetric. Thus $x_1 \succ x_3$, which is transitivity.

[3] If $x_1 \succ x_2, \cdots, x_{N-1} \succ x_N$, then by transitivity $x_1 \succ x_N$. Since $\succ$ is irreflexive, this implies $x_1 \neq x_N$. Thus $\succ$ is acyclic.

Preference Relations

When we are given a binary relation $\succ$ that expresses strict preference, we use it to define two other binary relations:

$$x_1 \succsim x_2 \qquad \text{if} \qquad x_2 \not\succ x_1$$

and

$$x_1 \sim x_2 \qquad \text{if} \qquad x_1 \not\succ x_2 \quad \text{and} \quad x_2 \not\succ x_1$$

where we are using $\not\succ$ as shorthand for "not $\succ$". The relation $\succsim$ is called *preference*, although it really expresses the absence of strict preference. The preference $\sim$ is called *indifference* — it expresses the absence of strict preference in either direction, which is perhaps not quite the same thing as active indifference.

Preference Relations

[Proposition 2.] If $\succ$ is a strict preference relation, then

(1) For all x_1 and x_2, exactly one of $x_1 \succ x_2$, $x_2 \succ x_1$ or $x_1 \sim x_2$ holds.

(2) $\succsim$ is complete and transitive.

(3) $\sim$ is reflexive, symmetric, and transitive.

(4) $x_1 \succ x_2$, $x_2 \sim x_3$, $x_3 \succ x_4$ imply $x_1 \succ x_3$ and $x_2 \succ x_4$.

(5) $x_1 \succsim x_2$ if and only if $x_1 \succ x_2$ or $x_1 \sim x_2$.

(6) $x_1 \succsim x_2$ and $x_2 \succsim x_1$ imply $x_1 \sim x_2$.

Preference Relations

Proof. [1] follows from the definition of $\sim$ and the fact that $\succ$ is asymmetric.

[2] By the asymmetry of $\succ$, either $x_1 \not\succ x_2$ and $x_2 \not\succ x_1$ (or both) for all x_1 and x_2, thus $\succsim$ is complete. For the transitivity of $\succsim$, note that this follows immediately the negative transitivity of $\succ$.

[3] $\sim$ is reflexive because $\succ$ is irreflexive. $\sim$ is symmetric because the definition of $\sim$ is symmetric. For the transitivity, suppose $x_1 \sim x_2 \sim x_3$. Then $x_1 \not\succ x_2 \not\succ x_3$ and $x_3 \not\succ x_2 \not\succ x_1$. By negative transitivity of $\succ$, $x_1 \not\succ x_3 \not\succ x_1$, or $x_1 \sim x_3$.

[4] If $x_1 \succ x_2 \sim x_3$, then by [1] one of $x_1 \succ x_3$ or $x_3 \sim x_1$ or $x_1 \succ x_3$. But $x_3 \succ x_1$ is impossible, since then transitivity of $\succ$ would imply $x_3 \succ x_2$. And $x_3 \sim x_1$ is impossible, since then transitivity of $\sim$ would imply $x_2 \sim x_1$, contradicting $x_1 \succ x_2$. Thus $x_1 \succ x_3$ must hold.

If $x_2 \sim x_3 \succ x_4$, then by [1] one of $x_2 \succ x_4$ or $x_4 \sim x_2$ or $x_2 \succ x_4$. But $x_4 \succ x_2$ is impossible, since then transitivity of $\succ$ would imply $x_3 \succ x_2$. And $x_4 \sim x_2$ is impossible, since then transitivity of $\sim$ would imply $x_4 \sim x_3$, contradicting $x_3 \succ x_4$. Thus $x_2 \succ x_4$ must hold.

[5] $x_1 \succsim x_2$ if and only if $x_2 \not\succ x_1$ if and only if $x_1 \succ x_2$ or $x_1 \sim x_2$ (by part [1]).

[6] This is immediate from the definition of $\succsim$ and $\sim$.

Preference Relations

Note well the plot: the agent expresses strict preferences, from which *we* define preferences and indifference. It is strict preference that is basic — the agent is not being called upon to express any judgments concerning weak preference or indifference, and he might disagree with our use of those terms to describe the negation of strict preference.

Another possible plot would be to ask the agent to express preferences or strict preference or indifference. That is, the basic relation is $\succsim$. That is a plot that is followed in many developments of choice theory, and in the standard treatment it leads to the same mathematical results.

Preference Relations

A binary relation $\succsim$ on a set S is called a preference relation if it is complete and transitive.

(1) $\succsim$ *is complete:* for all $x_1 \in S$ and $x_2 \in S$, $x_1 \succsim x_2$ or $x_2 \succsim x_1$ (or both).

(2) $\succsim$ *is transitive:* if $x_1 \succsim x_2$ and $x_2 \succsim x_3$, then $x_1 \succsim x_3$.

Preference Relations

When we are given a binary relation $\succsim$ that expresses preference, we use it to define two other binary relations:

$$x_1 \succ x_2 \qquad \text{if} \qquad x_2 \not\succsim x_1$$

and

$$x_1 \sim x_2 \qquad \text{if} \qquad x_1 \succsim x_2 \quad \text{and} \quad x_2 \succsim x_1$$

where we are using $\not\succsim$ as shorthand for "not $\succsim$". The relation $\succ$ is called *strict preference*, although it really expresses the absence of preference. The preference $\sim$ is called *indifference* — it expresses the preference in both directions.

Preference Relations

[Proposition 3.] If $\succsim$ is a preference relation, then

(1) For all x_1 and x_2, $x_1 \succsim x_2$ or $x_2 \succsim x_1$ (or both) holds.

(2) $\succ$ is asymmetric and negatively transitive.

(3) $\sim$ is reflexive, symmetric, and transitive.

(4) $x_1 \succ x_2$ if and only if $x_1 \succsim x_2$ but $x_2 \not\succsim x_1$.

(5) $x_1 \succ x_2$ if and only if $x_1 \succsim x_2$ but $x_2 \nsim x_1$.

Preference Relations

[Proposition 4.] Given a preference relation $\succsim'$ on a set S, define two new binary relation $\succ'$ and $\sim'$ from $\succsim'$ by

$$x_1 \succ' x_2 \qquad \text{if} \qquad x_2 \not\succsim' x_1$$

and

$$x_1 \sim' x_2 \qquad \text{if} \qquad x_1 \succsim' x_2 \quad \text{and} \quad x_2 \succsim' x_2.$$

Then if $\succsim'$ is complete and transitive, $\succ'$ will be a strict preference relation. Moreover, if we start with a binary relation $\succsim'$, define $\succ'$ and $\sim'$ as above from $\succsim'$, and then define $\succsim$ and $\sim$ from $\succ'$ by

$$x_1 \succsim x_2 \qquad \text{if} \qquad x_2 \not\succ' x_1$$

and

$$x_1 \sim x_2 \qquad \text{if} \qquad x_1 \not\succ' x_2 \quad \text{and} \quad x_2 \not\succ' x_1$$

then $\succsim'$ and $\succsim$ will agree, as will $\sim'$ and $\sim$.

Preference Relations

Proof. $\succ'$ is asymmetric because $x_1 \succ' x_2$ implies $x_2 \not\succsim' x_1$ from the definition of $\succ'$ and then $x_2 \not\succ' x_1$. $\succ'$ is negatively transitive because $x_1 \not\succ' x_2$ and $x_2 \not\succ' x_3$ implies $x_3 \succsim' x_2$ and $x_2 \succsim' x_1$ and then $x_3 \succsim' x_1$ from the completeness and transitivity of $\succsim'$, thus $x_1 \not\succ' x_3$ from the definition of $\succ'$.

$x_1 \succsim' x_2$ implies $x_2 \not\succ' x_1$ from the completeness of $\succsim'$, then $x_1 \succsim x_2$ from the definition of $\succsim$. $x_1 \succsim x_2$ implies $x_2 \not\succ' x_1$ from the definition of $\succsim$, then $x_1 \succsim' x_2$ from the completeness of $\succsim'$.

The proof of that $\sim'$ and $\sim$ are equivalent is from the definitions of the indifferences.

Preference Relations

So it doesn't matter whether we start with a strict preference relation that is asymmetric and negatively transitive or with a preference relation that is complete and transitive — we end up to in the same place. For reasons of interpretation we prefer to take strict preference as being basic. But it is a matter of personal taste, and most authors do it the other way.